MOHAMED AMINE AROUS

AI Engineer | Generative AI, LLMs, RAG & AI Agents

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Google Scholar

AI Engineer specializing in Generative AI, with experience developing and deploying solutions based on Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, and Knowledge Graphs. Experience at CEA List, École Polytechnique, and STMicroelectronics. Skilled in Python, FastAPI, AWS, Docker, and AI system evaluation, with a particular focus on reliability and production deployment.

Professional Experience

CEA List (LASTI Team) – Research Intern | AI Engineering & Knowledge Graphs **Apr 2026 – Oct 2026**
Supervisor: Prof. Gaël De Chalendar *Palaiseau, France*

- Developed a *Neuro-Symbolic AI* pipeline combining *Foundation Models* and *Knowledge Graphs*, achieving *92% enrichment accuracy* and improving *Hit@1 by 10%*.
- Benchmarked ML, Deep Learning, LLMs, and System 1 models for incorrect knowledge detection, preventing *250+ unsafe robot plans*.

CMAP Laboratory, École Polytechnique – Research Intern **May – Sep 2025**
Supervisor: Prof. Eric Moulines *Palaiseau, France*

- Developed *CosyVoice2-EU*, an open-source adaptation of a *Transformer*-based *voice cloning* model for European languages (GitHub).
- Achieved performance *exceeding the best open-source baselines* while remaining *competitive with a commercial speech synthesis baseline (ElevenLabs)*, contributing to a *paper accepted at EUSIPCO*.

Projects

Portfolio with RAG-Powered AI Assistant **2026**
Frontend GitHub *Backend GitHub* *Live*

- Developed a *serverless* assistant widget for my portfolio using *FastAPI* on *AWS Lambda* and hybrid *BM25/embeddings* search to answer visitors' questions based on my career data.
- Designed for near-zero operating costs and validated with *413 passing tests*, with zero hallucinations across 18 evaluation questions.

NORMA – Knowledge Graph & GraphRAG for Automotive Regulations GitHub **2026**
Normative Ontology for Requirements Modeling & Analysis *Live*

- Developed a bilingual *knowledge graph* pipeline transforming *UNECE/EU* regulations into a versioned, traceable *Neo4j* graph using an *OWL* ontology; *SHACL* validation reduced 21,700 violations to *0*.
- Deterministic link expansion improved multi-hop *Recall@10* by *2.2*$\times$ (0.317 $\rightarrow$ 0.708). Deployed on *AWS Lambda*, ~$0.003/month.

AWS CI/CD Deployment Pipeline GitHub **2026**

- Designed and deployed a fully automated *CI/CD pipeline* on *AWS*, taking a web application from Git commit to production without any manual steps.
- Automated artifact management, builds, deployment, and release orchestration using *CodePipeline*, *CodeBuild*, *CodeDeploy*, and *CodeArtifact*.
- Deployed to production on *EC2*: releases secured by *IAM*, triggered on every GitHub push.

Embodied AI with Vision-Language-Action Models (VLAs) GitHub **Oct 2025 – Mar 2026**
Academic project, team of 5, supervised by Prof. Gianni FRANCHI

- Developed an *automated benchmarking framework* to evaluate the AIUTA agent on *multiple LLM backbones* (*Qwen3-8B*, *Llama3-8B*) on a *GPU cluster*.

Multi-Agent Recruitment Assistant GitHub **2025**

Veo Hackathon | *Team of 6*

- Developed *AI agents* with *CrewAI* and *MistralAI* to automate candidate screening through *SWOT analysis* of CVs and evaluation of *body language* in video interviews.
- Automated candidate communication and report delivery using *Python* and *SMTP* email services.

Team Engineering Project **Oct 2024 – May 2025**

STMicroelectronics *Team project*

- Developed *machine learning models* in *Python* (*Scikit-learn*, *PyTorch*, Pandas, NumPy) to evaluate similarity between microelectronic components.
- Achieved *98% accuracy*, *outperforming the company's internal baseline solution*.
- Collaborated within a multidisciplinary engineering team to design and validate the final solution.

Education

Engineering Degree, Artificial Intelligence Specialization (Double Degree)

ENSTA Paris – ENIT 2023 – 2026

2nd year: GPA *3.96/4.3* | *1st year*: *Ranked 2*/50 (ENIT)

Coursework: Deep Learning, Machine Learning, Language Modeling, Reinforcement Learning, game theory & multi-agent control, probabilistic reasoning

Preparatory Classes in Mathematics and Physics

Institut Préparatoire aux Études d'Ingénieurs de Tunis (IPEIT) 2021 – 2023

Ranked *TOP 4*/177 (2nd year) | *TOP 3*/326 (1st year)

Coursework: linear algebra, probability, statistics, numerical analysis, physics, chemistry, Python programming, databases (SQL)

Technical Skills

- **Programming Languages:** Python, C++, SQL
- **Generative AI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Agentic AI, Transformers, Vector Embeddings, Semantic Search, Ollama, CrewAI
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, CUDA, Model Evaluation
- **Backend & APIs:** FastAPI, REST APIs, Flask, Streamlit, Gradio
- **Data & Semantic AI:** PostgreSQL, Neo4j, DynamoDB, Knowledge Graphs, BM25, RDF, OWL, SHACL, Named Entity Recognition (NER)
- **Cloud & DevOps:** AWS, AWS Lambda, EC2, IAM, CodePipeline, CodeBuild, CodeDeploy, CodeArtifact, Docker, CI/CD, Git, GitHub

Publications

- *"Europeanizing Modular Zero-Shot TTS: Component-Level Adaptation Framework for French & German."* EUSIPCO 2026. PDF Code

Awards & Distinctions

Excellence Scholarship (Tunisian Government) · **Google MENA ML 2026** (full scholarship, Google DeepMind) · **School Ambassador (5x)** (competitive programming & AI)

Certifications (all certifications)

- **AWS Certified AI Practitioner** – *Amazon Web Services*, 2026 Verification
- **Generative AI Engineering Professional Certificate** – *IBM*, 2025 (16-course program) Verification
- **Data Science Professional Certificate** – *IBM*, 2025 (6-course program) Verification
- **Foundations of AI Professional Certificate** – *IBM*, 2025 (3-course program) Verification

Volunteering & Leadership

Tutor – #GENIUS Program, École Polytechnique — weekly academic support in science & mathematics · **Competitive Programming Lead**, G^{2}FOSS — organized *DSA* courses and competitions

Languages

English (Linguaskill: **C1**) | French (TCF: **C1**) | Arabic (**Native**)